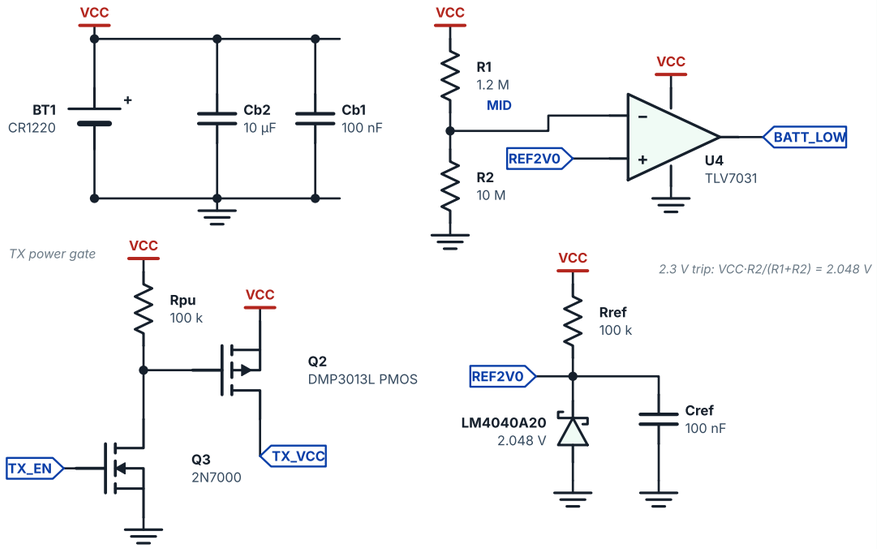
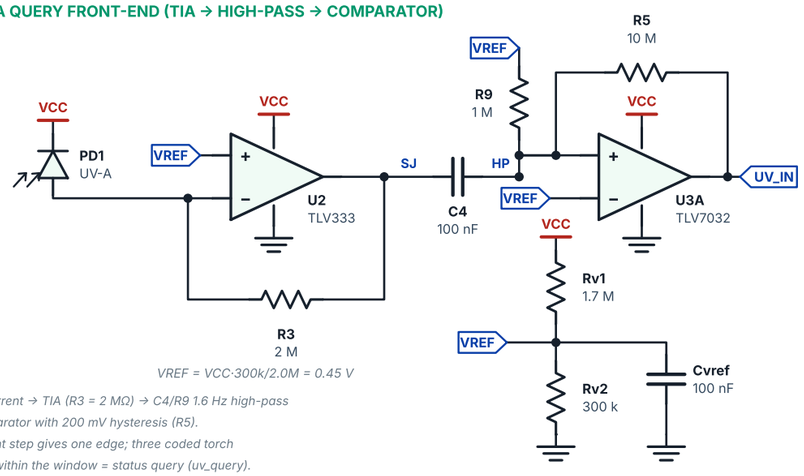
PULSE-Box Tag — Board Schematic

Sheet 1/2 · analog front-end, power, MCU and 433.92 MHz FSK transmitter · Rev A

1 · POWER · BATTERY MONITOR · TX SWITCH

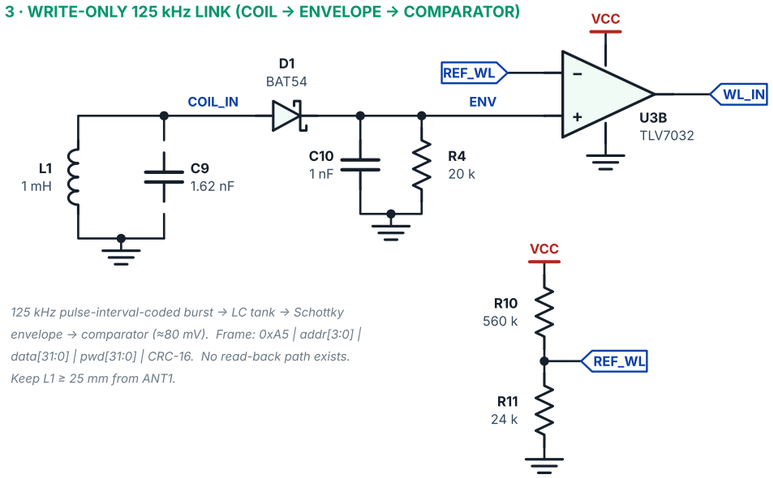


2 · UV-A QUERY FRONT-END (TIA → HIGH-PASS → COMPARATOR)



Photocurrent → TIA (R3 = 2 MΩ) → C4/R9 1.6 Hz high-pass → comparator with 200 mV hysteresis (R5). A sunlight step gives one edge; three coded torch flashes within the window = status query (uv_query).

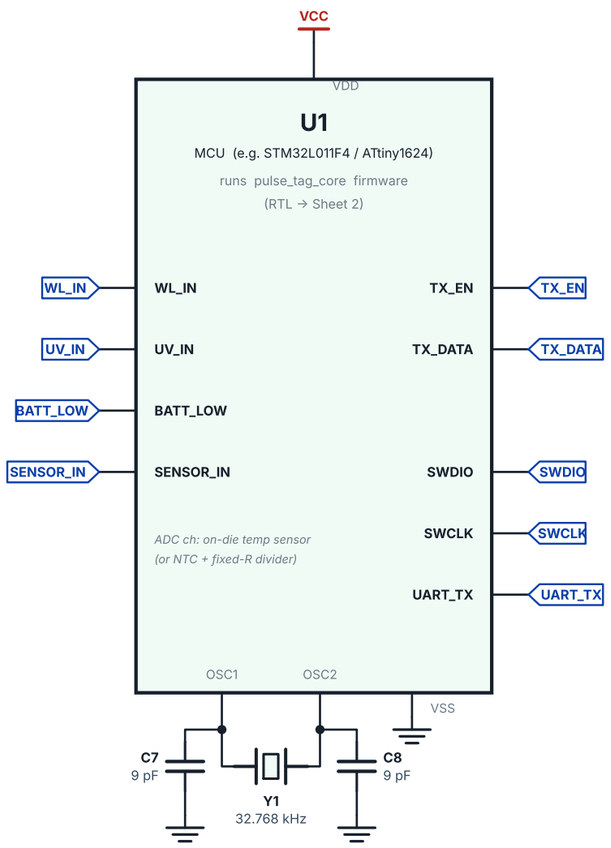
3 · WRITE-ONLY 125 kHz LINK (COIL → ENVELOPE → COMPARATOR)



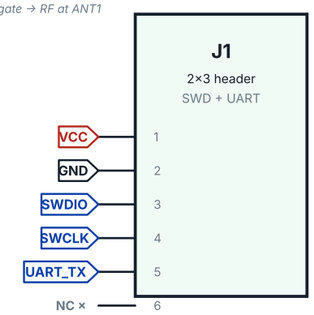
125 kHz pulse-interval-coded burst → LC tank → Schottky envelope → comparator (≈80 mV). Frame: 0xA5 | addr[3:0] | data[31:0] | pwd[31:0] | CRC-16. No read-back path exists. Keep L1 ≥ 25 mm from ANT1.

4 · MCU · CLOCK · PROGRAMMING

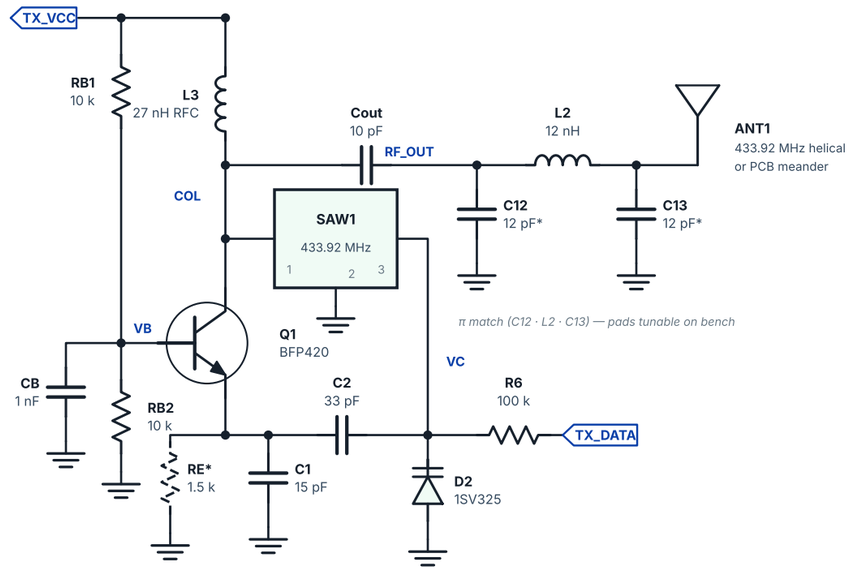
Cb1 100 nF / Cb2 10 µF decouple VDD (see 1)



Programming / bring-up:
power rails → J1 flash → 32.768 kHz at OSC1 → TX_DATA toggle → TX_VCC gate → RF at ANT1



5 · 433.92 MHz FSK TRANSMITTER (SAW-STABILISED COMMON-BASE COLPITTS)



TX_EN → Q3 → Q2 gates TX_VCC (radio is unpowered between packets).
TX_DATA (0/VCC) through R6 pulls varactor D2 → ±36 kHz FSK shift.
Emitter node needs a DC return: RE (dashed) is NOT in the BOM — see review note R2.

TX link budget (from RTL)

433.92 MHz · 2-FSK ±36 kHz · NRZ 4096 bit/s (32.768 kHz ÷ 8)
96-bit frame: preamble 0xAA (8) | sync 0x2DD4 (16) | box ID (32) | status (8) | sample (8) | seq (8) | CRC-16/CCITT (16)
Heartbeat 6 h + jitter; alert back-off 10 s → 300 s; UV query on demand

** = simulated / bench-tuned value*

DESIGN & REVIEW NOTES

Design

1. Pilot option: replace Q1/SAW1/D2 with a sub-GHz SoC module (E22-900M / STM32WL): TX_DATA→DIN, TX_EN→sleep.
2. UV photodiode needs a clear optical window in the label / enclosure.
3. Solid ground plane under RF; keep L1 ≥ 25 mm from ANT1; π-match pads left tunable.

Review flags (vs. original package)

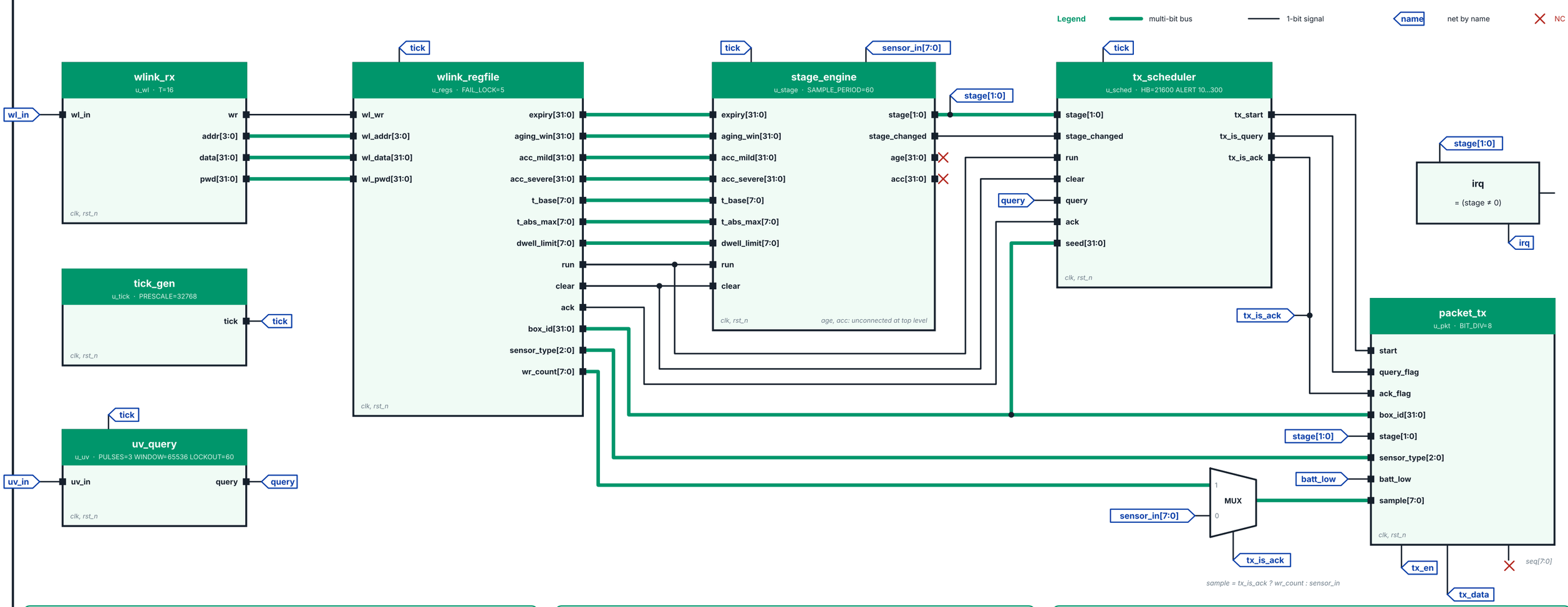
- **R1** BOM/netlist put Q2 source on TX_VCC; drawn here as source = VCC, drain = TX_VCC (correct high-side switch).
- **R2** Q1 emitter has no DC path to GND in the BOM → RE ≈ 1–2.2 kΩ added (dashed). Confirm in simulation.
- **R3** R10/R11 on VCC give 3.0·24/584 ≈ 123 mV, not 80 mV. For 80 mV use R10 ≈ 866 kΩ.
- **R4** J1 pin 6 not defined in the source (drawn NC).

PULSE-Box — Reusable Shelf-Life Monitoring Tag

Board Schematic · power, sensing, MCU, RF	Team VK26-IITI-AHM-0520
Sheet 1 of 2 · Format A3	Vishwakarma Awards 2026-27 · Stage 2
Rev A · 2026-10-16	Source: netlist.csv, BOM.csv, RTL
Passives 0603 unless noted · * = tuned value	Red = power · blue = net name

PULSE-Box Tag — Digital Core (pulse_tag_core.v)

Sheet 2/2 · RTL block schematic: every instance, port and net from the Verilog · Rev A



REGISTER MAP (WRITE-ONLY, PASSWORD-GATED)

addr	register	notes
0	box_id[31:0]	also seeds the TX jitter LFSR
1	expiry[31:0]	seconds → CRITICAL when age ≥ expiry
2	aging_win[31:0]	default 7 776 000 s (90 d) → AGING
3	acc_mild[31:0]	AGING when Σ excess ≥ value
4	acc_severe[31:0]	CRITICAL when Σ excess ≥ value
5	cfg[31:0]	t_base [7:0] · t_abs_max [15:8]
		sensor_type [18:16] · dwell_limit [27:20]
6	password[31:0]	every frame must carry it
8	cmd	[0] start: run=1, clear, wr_count++ · [1] stop

Wrong password → frame dropped, link locked FAIL_LOCK = 5 s. No read path exists.

Reset: aging_win = 90 d; thresholds = 0xFFFFFFFF (disabled); run = 0.

STAGE ENGINE & TX SCHEDULE

- stage 0 = FRESH · 1 = AGING · 2 = CRITICAL; only moves up; cleared only by an authenticated restart.
- Every SAMPLE_PERIOD = 60 ticks (1 Hz): acc += max(sensor_in − t_base, 0); dwell++ while sensor_in ≥ t_abs_max.
- CRITICAL: age ≥ expiry │ acc ≥ acc_severe │ dwell ≥ dwell_limit.
- AGING: age + aging_win ≥ expiry │ acc ≥ acc_mild.
- FRESH: heartbeat every 21 600 s (+ LFSR jitter ≤ 255 s).
- AGING/CRIT: immediate packet on change, then back-off 10 s → 300 s (×2, jitter ≤ 7 s).
- UV query: 3 rising edges within 65 536 clk (2 s) → 1 packet; 60 s lock-out.
- ack packet after every valid write; its 'sample' byte carries wr_count.

FRAME FORMATS

Write link · coil → tag · 92 bit, pulse-interval coded (T = 16 clk → 0, 2T → 1)

HDR A5	ADR	DATA	PWD	CRC16
8	4	32	32	16

Uplink · tag → receiver · 96 bit, MSB first, NRZ, 4096 bit/s

PRE AA	SYNC 2DD4	BOX ID	STATUS	SAMPLE	SEQ	CRC16
8	16	32	8	8	8	16

STATUS = { stage[1:0], sensor_type[2:0], batt_low, query_flag, ack_flag }

CRC-16/CCITT-FALSE (poly 0x1021, init 0xFFFF): addr·data·pwd (write) · 7 payload bytes (uplink).

Idle gap > 4T aborts a write frame; every frame is self-contained (stateless).

Uplink bit time = 8 clk of 32.768 kHz = 244 µs → 96 bits ≈ 23.4 ms on air.

Top-level ports

in : clk · rst_n · wl_in · uv_in · sensor_in[7:0] · batt_low

out: tx_en · tx_data → FSK modulator · irq = (stage ≠ 0)

RTL verified in simulation: 18 / 18 testbench checks (Icarus Verilog).

Drawn from: pulse_tag_core · tick_gen · uv_query · wlink_rx · wlink_regfile · stage_engine · tx_scheduler · packet_tx (.v)

PULSE-Box — Reusable Shelf-Life Monitoring Tag

Digital core · RTL block schematic

Sheet 2 of 2 · Format A3

Rev A · 2026-10-16

Team VK26-IITI-AHM-0520

Vishwakarma Awards 2026-27 · Stage 2

Companion: Sheet 1 (board circuit)